FPT ACADEMY INTERNATIONAL

FPT – APTECH COMPUTER EDUCATION

InterVue

Supervisor: *HOÀNG ĐỨC QUANG*

Semester: *IV*

Batch No: *T2.2410.E0*

Group No: *01*

Order:	**Full name**	**Roll No.**
1.	*Phạm Hoàng Tuấn*	*Student1614869*
2.	*Mai Trung Hậu*	*Student1604660*
3.	*Lê Minh Trí*	*Student1604679*
4.	*Lâm Hoàng An*	*Student1614870*

***Month**: 10 **Year**: 2026*

This is to certify that

Mr. **PHẠM HOÀNG TUẤN**

Mr. **MAI TRUNG HẬU**

Mr. **LÊ MINH TRÍ**

Mr. **LÂM HOÀNG AN**

Have successfully Designed & Developed

InterVue

Submitted by:

Date Of Issue:

Authorized Signature:

Content

InterVue **1**

Content **3**

Acknowledge **4**

Synopsis **8**

Review 01 ..... **14**

Functional Requirements **15**

I. FUNCTIONAL REQUIREMENTS 15

System Requirements **25**

II. SYSTEM REQUIREMENTS 25

1. Server requirements 25

2. Client requirements 26

3. Development Software 27

Task Sheet – Review 1 29

III. TASK SHEET - REVIEW 1 ..... **ERROR! BOOKMARK NOT DEFINED.**

Acknowledgments and Declaration **31**

Acknowledge

Every software project begins with a simple question. Ours was this: *why is it still so hard for a talented developer and a company that needs them to find each other and trust each other?* InterVue grew out of that question. As we bring this project to completion, we feel a deep sense of accomplishment, and an even deeper sense of gratitude to everyone who helped turn that question into a working platform.

InterVue was never meant to be another job board. We set out to build a recruitment platform dedicated to the Information Technology industry, where every job post states its salary, every business is verified before it reaches candidates, every candidate confirms their identity before applying, and every step of the hiring process can be seen and followed. Translating that vision into software turned out to be far more demanding than we first imagined, and far more educational.

A Journey Through Real-World Complexity

This project placed us in front of problems that rarely appear in textbooks, but appear in every real product.

We learned that **trust has to be engineered**. Building electronic identity verification with FPT.AI meant much more than calling an API. We had to decide what happens when a confidence score falls into the uncertain range, when the same citizen ID appears on two accounts, or when a user fails verification five times. On the employer side, we designed a three-level verification process, from email to business license to domain verification, so that candidates can tell a genuine company from a fraudulent one at a glance.

We learned that **money demands precision**. Integrating bank transfers through SePay taught us to think in terms of orders, expiry windows, idempotent processing, and exact reconciliation. A single transaction must never be credited twice, and an

underpayment must never silently activate a service. These details may be invisible to users, but they decide whether a platform deserves its users' trust.

We learned that **security is a series of careful decisions**. Generic login error messages, CAPTCHA after repeated failures, two-factor authentication with recovery codes, single-use links that expire in minutes, and a seven-day grace period before an account is permanently deleted are not glamorous features. Each one, however, reflects a decision about how to protect people's accounts and personal data.

We learned that **real-time systems change how people communicate**. Implementing chat over WebSocket, interview scheduling with reminders, and a clear application timeline showed us how much a candidate's experience depends on knowing where they stand. A simple "Viewed" status, with the time a recruiter opened their CV, can mean a great deal to someone waiting for an answer.

Most importantly, we learned that **artificial intelligence must be used responsibly, especially when people's careers are at stake**. Throughout InterVue, AI evaluates CVs, simulates interviews, recommends jobs, drafts job descriptions, and answers usage questions through a retrieval-augmented chatbot. From the very beginning, however, we committed to a clear principle: *AI assists, humans decide*. Every AI assessment must point to evidence in the CV. AI must never invent experience or skills a candidate does not have. It must never use age, gender, photos, hometown, or identity data to judge a person. A missing detail in a CV does not mean a missing skill. Holding ourselves to these principles made us think not only as developers, but as people responsible for the impact of what we build.

Each of these challenges forced us to slow down, ask harder questions, and accept that good software is defined as much by its rules, limits, and edge cases as by its visible features.

Heartfelt Gratitude to Our Supervisor and FPT Aptech

First and foremost, we would like to express our sincere gratitude to **FPT Aptech** for giving us the opportunity to undertake this project. The project-based learning

approach allowed us to experience the full lifecycle of software development, from problem definition and requirement analysis to system design, implementation, and testing, in a way that no lecture alone could provide.

We are especially grateful to our supervisor, **Mr. HOÀNG ĐỨC QUANG**. His guidance has been the compass of this project. He taught us to define requirements precisely before writing a single line of code, to question our own assumptions, and to examine every feature from the perspective of each party who would use it: the candidate, the recruiter, and the administrator. His thoughtful feedback during each review pointed out weaknesses we had overlooked and pushed our work to a higher standard. Beyond technical knowledge, he showed us the mindset of a professional engineer: careful, honest, and always focused on the real problem. Without his patience and dedication, InterVue would not have reached the level it has today.

A Salute to Our Team

This project belongs to the four of us: **Phạm Hoàng Tuấn, Mai Trung Hậu, Lê Minh Trí, and Lâm Hoàng An**. Along the way we shared long discussions over requirements, late nights of debugging, disagreements that were settled by better arguments rather than louder voices, and the quiet satisfaction of watching separate modules finally come together as one system.

Working together taught us lessons that go beyond code. We learned to divide responsibilities clearly, to review each other's work with honesty and respect, to communicate when something was going wrong, and to hold the whole system to a shared standard rather than to individual comfort. Each member brought different strengths, and InterVue is stronger because of all of them. We are proud of what we built together, and even prouder of how we grew as a team in the process.

A Heartfelt Thanks to Our Families and Friends

We also owe deep thanks to our families and friends. They patiently accepted our long hours in front of screens, our missed gatherings, and our constant talk of deadlines and bugs. Their encouragement kept us going through the most demanding

stages of this project, and their belief in us gave us the confidence to aim higher. Though their names do not appear in our source code, their support is present in every part of this work.

A Project That Transcends the Classroom

For us, InterVue is far more than an academic requirement. It reflects a problem we genuinely care about. Many young developers, interns, and fresh graduates struggle to find opportunities they can trust, while many companies struggle to identify the right people among hundreds of applications. Through InterVue, we have tried to show that technology can make this process clearer, fairer, and more humane: salaries are transparent, businesses and candidates are verified, progress can be tracked at every stage, and AI supports human judgment instead of replacing it.

Looking Forward to New Horizons

As we close this chapter, we carry with us much more than a finished project. We carry the experience of analyzing real requirements, designing systems that must handle failure gracefully, protecting users' data and money, and using artificial intelligence with care and accountability. These lessons will remain with us long after this semester ends, and they will guide us as we step into the IT industry as developers, problem-solvers, and responsible builders of technology.

With heartfelt thanks and great excitement for the journey ahead,

Team 01

Synopsis

InterVue is a recruitment platform dedicated to the Information Technology industry. It connects IT job seekers, from interns and fresh graduates to senior engineers, with businesses that need technical talent. The platform covers the entire hiring journey in one place: publishing job posts, discovering opportunities, submitting applications, screening candidates with the support of artificial intelligence, scheduling interviews, communicating in real time, and recording the final hiring outcome. Its defining principle is simple: *AI assists, humans decide.*

Background and Motivation

The IT job market in Vietnam is growing rapidly, yet the hiring experience for both sides remains frustrating. Candidates frequently encounter job posts with vague requirements, hidden salaries, or fraudulent intentions, including requests to pay deposits or fees before being considered. Once they apply, many receive no feedback at all and have no way of knowing whether their CV has even been opened. Recruiters, on the other hand, spend a large amount of time manually reading hundreds of applications, many of which do not match the position, and they have few reliable tools to compare a CV against the actual technical requirements of a job. On general-purpose platforms, there is often little to confirm that a candidate is who they claim to be, or that a company posting a job is a genuine, legally registered business.

InterVue addresses these problems through four commitments: **transparency** of job information, **trust** between both parties, **efficiency** in evaluating applications, and **visibility** of the application process.

Users and Roles

The system serves three roles.

Candidates are people looking for full-time, part-time, or internship positions in IT. **Recruiters** are representatives of businesses or household businesses who publish job posts and manage hiring; individuals cannot post jobs on InterVue. **Administrators** operate the platform: they verify identities and businesses, moderate job posts and reports, manage shared catalogs and service packages, handle transactions and refunds, and respond to support requests.

Candidate and recruiter accounts are maintained through separate portals. The same email address may be used to register on both, but each account has its own credentials, profile, verification status, service packages, and transaction history. A recruiter can never apply to a job published under their own account.

Building Trust on Both Sides

Trust is the foundation of InterVue, and it is enforced through verification rather than assumed.

Before submitting any application, a candidate must complete electronic identity verification (eKYC) through FPT.AI. The process consists of capturing both sides of the citizen identity card and performing a liveness face check. High-confidence results are approved automatically. Uncertain results, duplicate identity numbers, or repeated failures are escalated to an administrator for manual review. The minimum age of 16 is confirmed from the date of birth on the identity card.

Recruiters progress through three verification levels. The first level requires a verified email. The second requires phone verification, complete company information matched against tax records, and an approved business or household-business license. The third level adds verification through an email address on the company's own domain. Each level unlocks additional capabilities, such as free job posts, CV views in candidate search, and a visible verification badge on job posts and company pages. Recruiters must also accept data processing agreements before they can receive any CV.

Every job post, without exception, is reviewed by an administrator before it becomes visible. AI pre-screens each post for warning signs such as requests for payment from candidates, discriminatory requirements, or abnormal salaries, and flagged posts are moved to the front of the review queue. Candidates can report suspicious posts. When three different verified candidates report the same post within 24 hours, it is automatically hidden until an administrator investigates.

Transparent Job Information

Every job on InterVue is paid; unpaid positions are not permitted. Each post states the position, level, required skills, working arrangement (on-site, hybrid, remote, or flexible), contract type, work location, and salary. The salary can be shown as a range, a minimum, a maximum, or as negotiable. For full-time and part-time positions, the system validates that the stated salary is not below the regional minimum wage defined by Vietnamese law.

Positions, levels, skills, and locations are drawn from standardized catalogs shared by both sides. Locations follow Vietnam's new two-level administrative structure of province or city and ward or commune. This shared vocabulary allows job search, recommendations, and AI matching to compare candidates and jobs consistently.

The Candidate Experience

Candidates can search for jobs by title, company, and location without signing in, and can explore jobs near them on a map within a chosen radius. After registering, a guided onboarding helps them upload a CV, let AI fill in their profile, confirm their desired position, and enable job alerts.

The candidate profile goes well beyond a single document. It includes work experience, standardized skills with proficiency levels and years of use, notable projects with links to source code and live demos, education, certificates, languages, and job expectations. Candidates control what recruiters can see. Birth year and gender are hidden by default and are never used by AI, and contact details are revealed only to

recruiters of jobs the candidate has applied to, or after the candidate accepts a chat invitation.

Once applications are submitted, candidates follow each one through a clear timeline of statuses: Submitted, Viewed, Under Review, Interview, Hired, Not Selected, Withdrawn, or Job Closed. They can see exactly when a recruiter first opened their CV. They can confirm, decline, or propose a new time for interviews, add them to Google Calendar, and receive reminders 24 hours and one hour in advance.

The Recruiter Experience

Recruiters organize their hiring into campaigns, where each campaign represents one position and contains its job post, applications, recommended candidates, active services, and reports. AI helps them draft job descriptions from a position and level, or fill in a post automatically from an existing job description file.

Recruiters can request that candidates answer screening questions, set expected answers, and have non-matching applications flagged rather than rejected. They manage applications through a structured pipeline, keep internal notes, select standardized rejection reasons, and choose whether to share those reasons with the candidate. Verified recruiters can also search for candidates who have chosen to be discoverable, view AI-recommended profiles for each campaign, and invite candidates to chat. Recruitment reports present the hiring funnel, from job views to applications, interviews, and hires, along with response rates and the effect of paid services.

Artificial Intelligence with Accountability

AI is integrated throughout InterVue, always as an assistant and never as a decision-maker.

For candidates, AI reviews CVs at three levels: a general review against common standards, a review against the candidate’s desired position, and a detailed fit check against a specific job post. It also provides text-based mock interviews with feedback on each answer, job recommendations that explain why each job fits, and cover letter

drafts based on the candidate's own CV. For recruiters, AI compares each CV against the job requirements, highlights matched and missing skills with evidence from the CV, classifies applications into high, medium, or low fit, and points out what should be clarified in an interview. A separate chatbot built on retrieval-augmented generation answers questions about how to use the platform, and only about the platform.

These features follow strict principles. Every assessment must be grounded in evidence from the CV, presented as the relevant passage, the issue, and a suggested fix. AI must never invent experience or skills the candidate does not have. Age, gender, photos, hometown, and identity data are never used in any evaluation. A missing detail in a CV is never interpreted as a missing skill. AI never rejects or hires anyone automatically, and users are informed before their data is processed by AI. If an AI request fails, no quota is deducted, and paid usage is refunded automatically.

Security, Payments, and Data Protection

InterVue supports registration with email or social accounts, enforces strong password rules, and requires CAPTCHA after repeated failed login attempts without locking accounts. Users can enable two-factor authentication through an authenticator app or email, keep recovery codes, review their active devices, and sign out of all devices at once.

Paid services, such as candidate plans, job-fit checks, promoted job posts, add-ons, and recruiter credits, are purchased through bank transfer via SePay using a QR code generated for each order. The system automatically matches incoming transactions to orders, processes each transaction exactly once, and activates the service only when the payment is sufficient.

Account deletion follows a seven-day grace period, after which personal data, identity verification data, CVs, and conversations are removed. Only transaction records required by accounting and tax regulations, and records related to violations, are retained.

Platform and Scope

InterVue is designed primarily as a responsive web application, which offers the most comfortable environment for reading and comparing CVs, managing job posts, and operating the system. Its long-term direction includes a mobile application built with Flutter for quick job discovery, application tracking, and notifications, sharing the same accounts, job posts, and applications as the web platform.

Conclusion

InterVue brings together verified identities, transparent job information, a visible application pipeline, and responsible artificial intelligence in a single platform built for the IT industry. For candidates, it offers opportunities they can trust and a clear view of where they stand. For recruiters, it offers faster and better-informed screening without giving up human judgment. For our team, it has been a practical exercise in the full software development lifecycle, from defining the problem and analyzing requirements to designing a secure, reliable, and fair system that addresses a real need in the market.

Review 01

Functional Requirements

I. Functional Requirements

This section describes what each type of user can do in InterVue. The system has four roles: guest, candidate, recruiter, and administrator. A guest is a visitor who has not signed in. Candidate and recruiter accounts are separate, even when they use the same email address, and a Super Admin creates administrator accounts.

1. Guest

1.1 Browsing jobs and companies

- A guest can search for jobs by job title, company name, or location.
- A guest can read the details of a job post.
- A guest can view nearby job posts on the "Jobs near you" map.
- A guest can choose a search radius on the map.
- A guest can view a company page and its current openings.
- A guest can view the featured companies on the homepage.
- A guest can read the FAQ and Blog pages.

1.2 Using the help chatbot

- A guest can ask the help chatbot questions about how to use InterVue.

1.3 Registering and signing in

- A guest can register a candidate account with an email address or a social media account.
- A guest can register a recruiter account with an email address or a social media account.
- A guest can sign in with an email address and password or a social media account.
- A guest can request a password reset link by entering the account email address.

2. Candidate

2.1 Account verification and security

- A candidate can verify an email address by opening the verification link sent to that address.
- A candidate can request a new verification email.
- A candidate can complete identity verification (eKYC) with a citizen identity card and a face check.
- A candidate can retry identity verification and can request a manual review.
- A candidate can enable two-factor authentication.
- A candidate can review the active devices and sign out of one device or of all devices.

- A candidate can link or unlink social accounts and can change the password or the email address.

2.2 Onboarding and profile

- A candidate can follow a guided onboarding to set up the profile, and can skip steps and resume them later.
- A candidate can view a dashboard that summarizes applications, interviews, recommended jobs, and profile status.
- A candidate can manage a profile with personal information, experience, skills, education, and job expectations.
- A candidate can control which personal information recruiters can see and whether recruiters can find the candidate.
- A candidate can review the profile changes that AI proposes from a newly uploaded CV and choose which ones to apply.

2.3 CV management

- A candidate can upload a CV as a PDF file.
- A candidate can rename a CV, set a default CV, and delete a CV.
- A candidate can store several CVs, depending on the service plan.

2.4 Job discovery

- A candidate can filter job search results by position, level, salary, and working arrangement.

- A candidate can save jobs and review them later.
- A candidate can receive AI job recommendations that explain why each job fits.
- A candidate can create daily or weekly job alerts.
- A candidate can follow companies and receive notifications about their new job posts.

2.5 AI support

- A candidate can request AI feedback on a CV in general, for the desired position, or for a specific job post.
- A candidate can practice text-based mock interviews and receive feedback on each answer and a summary.
- A candidate can review the history of past mock interview sessions.
- A candidate can use AI to draft a cover letter, then edit, save, and download it as a PDF file.

2.6 Applications

- A candidate can apply for a job with a CV and answers to screening questions.
- A candidate can follow the status of each application.
- A candidate can withdraw an application and can reapply later.

2.7 Interviews and chat

- A candidate can confirm, decline, or propose a new time for an interview.

- A candidate can add an interview to Google Calendar and receive reminders before it starts.
- A candidate can chat in real time with recruiters.
- A candidate can accept or decline a chat invitation from a recruiter.

2.8 Plans, support, and safety

- A candidate can purchase service plans and review the payment history.
- A candidate can submit support requests.
- A candidate can report suspicious job posts and can block companies.
- A candidate can request account deletion.

3. Recruiter

3.1 Registration and verification

- A recruiter can post jobs on behalf of a business or a household business.
- A recruiter can describe the current hiring needs to receive suitable service suggestions.
- A recruiter can verify an email address (Level 1), and can then prepare the account, create campaigns, and draft job posts.
- A recruiter can verify a phone number with an OTP.
- A recruiter can complete the company profile and can submit a business license for approval.

- A recruiter at Level 2 can post jobs for free, view CVs in candidate search, and display a verification badge.
- A recruiter can verify an email address on the company's own domain (Level 3) and receive a high-level verification badge.
- A recruiter can accept the data processing agreements.
- A recruiter can enable two-factor authentication.

3.2 Company management

- A recruiter can select an existing company or create a new company.
- A recruiter can submit a letter of introduction or authorization to join an existing company.
- A recruiter can request changes to locked company information with supporting documents.

3.3 Campaigns and job posts

- A recruiter can view a dashboard that summarizes campaigns, job posts, applications, interviews, services, and warnings.
- A recruiter can create a recruitment campaign for each position.
- A recruiter can create a job post with structured information such as position, level, skills, and work location.
- A recruiter can state a salary for each job post.

- A recruiter can use AI to draft a job description or to fill in a job post from an existing description file.
- A recruiter can configure screening questions with expected answers.
- A recruiter can submit job posts for review and can edit, extend, pause, reopen, and close them.
- A recruiter can edit a live job post while the previously approved version stays visible until the change is approved.

3.4 Application management

- A recruiter can filter, search, and sort the applications of a campaign.
- A recruiter can view CVs, screening answers, and the AI fit assessment of each application.
- A recruiter can change the status of an application and record a rejection reason.
- A recruiter can keep internal notes on an application.
- A recruiter can save promising candidates and can report suspicious candidates.

3.5 Candidate search

- A recruiter can search for candidates who have chosen to be discoverable.
- A recruiter can view the AI-recommended profiles of each campaign by using CV view quotas or credits.
- A recruiter can send a chat invitation to a candidate.

3.6 Interviews and chat

- A recruiter can create an interview schedule for an application.
- A recruiter can accept a new time proposed by a candidate or propose another time.
- A recruiter can chat in real time with candidates.

3.7 Services, payments, and reports

- A recruiter can purchase service packages, add-ons, and credits.
- A recruiter can activate a purchased service on a job post.
- A recruiter can request a VAT invoice for a payment.
- A recruiter can review the history of services, credits, and account changes.
- A recruiter can view recruitment reports on the funnel, processing time, candidate sources, and service effectiveness.

4. Administrator

4.1 Access and overview

- A Super Admin can create administrator accounts.
- An administrator can sign in with two-factor authentication.
- An administrator can access the modules of the assigned internal sub-role.
- An administrator can view an overview dashboard of users, pending verifications, pending job posts, reports, and revenue.

- An administrator can view the audit log of administrator actions.

4.2 Accounts and verification

- An administrator can search for, view, lock, and unlock candidate and recruiter accounts.
- An administrator can reset the two-factor authentication of a user.
- An administrator can review escalated eKYC cases and can approve, reject, or grant additional attempts.
- An administrator can review business verification requests and company information changes, and can approve, reject, or request additional documents.

4.3 Content moderation

- An administrator can review job posts in a prioritized queue.
- An administrator can compare an edited job post with its approved version.
- An administrator can approve, reject, or hide job posts.
- An administrator can process reports on job posts, candidates, and conversations, and can inform the reporter of the outcome.

4.4 Catalogs, services, and finance

- An administrator can maintain the catalogs of positions, levels, and skills.
- An administrator can maintain the administrative units and the regional minimum wage table.

- An administrator can configure service plans, packages, add-ons, and credits.
- An administrator can monitor orders and transactions and can reconcile them manually when needed.
- An administrator can process refunds and VAT invoice requests and can export revenue reports.
- An administrator can generate, export, track, and revoke batches of Education plan codes.

4.5 Support and content

- An administrator can handle support tickets and consultation requests.
- An administrator can manage the FAQ, the Blog, the featured companies, and the knowledge base of the help chatbot.
- An administrator can send announcements to candidates, recruiters, or all users.

System Requirements

II. System Requirements

1. Server requirements

1.1 Hardware

Component	Requirement
CPU	**Processor type:** 64-bit (x86_64) multi-core processor **Processor cores:** Minimum: 2 cores · Recommended: 4 cores or more **Processor speed:** Recommended: 2.5 GHz or faster
OS	Ubuntu Server 22.04 LTS or higher (recommended) Windows Server 2019 or higher
Memory (RAM)	**Minimum:** 4 GB **Recommended:** 8 GB or more
Hard Drive	**Free space:** Minimum: 20 GB (SSD) · Recommended: 100 GB SSD or more **Maximum:** Operating system maximum
Network	Static public IP or domain name, with HTTPS (SSL certificate)

1.2 Software

Component	Requirement
Java Runtime	JDK 25
Backend Framework	Spring Boot (compatible with JDK 25)
Frontend Runtime	Node.js (LTS version supported by Next.js 16)
Frontend Framework	Next.js 16.x
RDBMS	MySQL 8.x
Cache / In-memory Store	Redis 7.x
Web Server / Reverse Proxy	Nginx (HTTPS, WebSocket support)
Third-party Services	FPT.AI (eKYC), SePay (payment), AI language model API, email delivery service

2. Client requirements

1.1 Hardware (Web)

Component	Requirement
CPU	**Processor type:** Intel Core i3 / AMD Ryzen 3 or equivalent **Processor speed:** Recommended: 2.0 GHz or faster
OS	Windows 10 or higher, macOS 12 or higher, Linux

Memory (RAM)	**Minimum:** 4 GB **Recommended:** 8 GB or more **Maximum:** Operating system maximum
Hard Drive	**Free space:** Minimum: 500 MB (browser cache and downloaded files)
Camera	Webcam or phone camera (required for identity verification)
Network	Stable Internet connection (minimum 5 Mbps)

1.2 Hardware (Mobile)

Component	**Requirement**
OS	Android 8.0 or higher
Memory (RAM)	**Minimum:** 3 GB **Recommended:** 4 GB or more
Storage	**Free space:** Minimum: 200 MB
Camera	Front and rear camera (required for identity verification)

Software (Mobile)

Component	**Requirement**
Mobile Application	InterVue mobile app (Flutter) for Android

3. Development Software

- IntelliJ IDEA (Backend – Spring Boot)

- Visual Studio Code (Frontend – Next.js, TypeScript, Tailwind CSS)
- Android Studio (Mobile – Flutter, Android build)
- JDK 25
- Node.js
- Flutter SDK
- MySQL 8.x and MySQL Workbench
- Redis
- Postman (API testing)
- Git and GitHub (version control)

III. Task Sheet – Review 1

S.No	Task	Deliverable	Assigned to	Status
1	Set up the project architecture, database, and security framework	Spring Boot and Next.js project base, database schema, JWT role-based authorization	Phạm Hoàng Tuấn	In Progress
2	Develop the AI module	AI CV evaluation, CV feedback, profile auto-fill, job recommendation, and job description generation	Phạm Hoàng Tuấn	Not Started
3	Develop the payment and service module	SePay payment, order reconciliation, service packages, and credit management	Phạm Hoàng Tuấn	Not Started
4	Integrate and deploy the system	Integrated system deployed on the server	Phạm Hoàng Tuấn	Not Started
5	Develop recruiter verification and company management	Phone verification, company profile, business license verification	Mai Trung Hậu	Not Started
6	Develop campaign and job post management	Campaign and job post CRUD, salary validation, screening questions, job post moderation	Mai Trung Hậu	Not Started
7	Develop application management and interview scheduling	Application pipeline, candidate search, interview schedules	Mai Trung Hậu	Not Started
8	Develop real-time chat and notifications	WebSocket chat, chat invitations, in-app and email notifications	Mai Trung Hậu	Not Started
9	Develop candidate profile and CV management	Candidate profile, CV upload and management, onboarding	Lâm Hoàng An	Not Started
10	Develop identity	eKYC integration and admin eKYC	Lâm Hoàng An	Not

	Prepared by (Student)	**Approved by (Faculty)**
	Project Group No: 01	hoa
Signature		
Date		

	verification (eKYC)	review		Started
11	Develop job search and job application	Job search, job map, saved jobs, job alerts, job application, application tracking	Lâm Hoàng An	Not Started
12	Develop authentication	Register, login, logout, email verification, password reset	Lê Minh Trí	In Progress
13	Develop guest pages	Home page, company page, FAQ, Blog	Lê Minh Trí	Not Started
14	Develop basic admin management	User management, catalog management, FAQ and Blog management	Lê Minh Trí	Not Started

Review 02

I. Architecture Design

1. Application and Technology Architecture

The technical architecture of InterVue is organized into four core layers, decoupling client experiences, centralized API gateway routing, core business logic, and transactional data persistence.

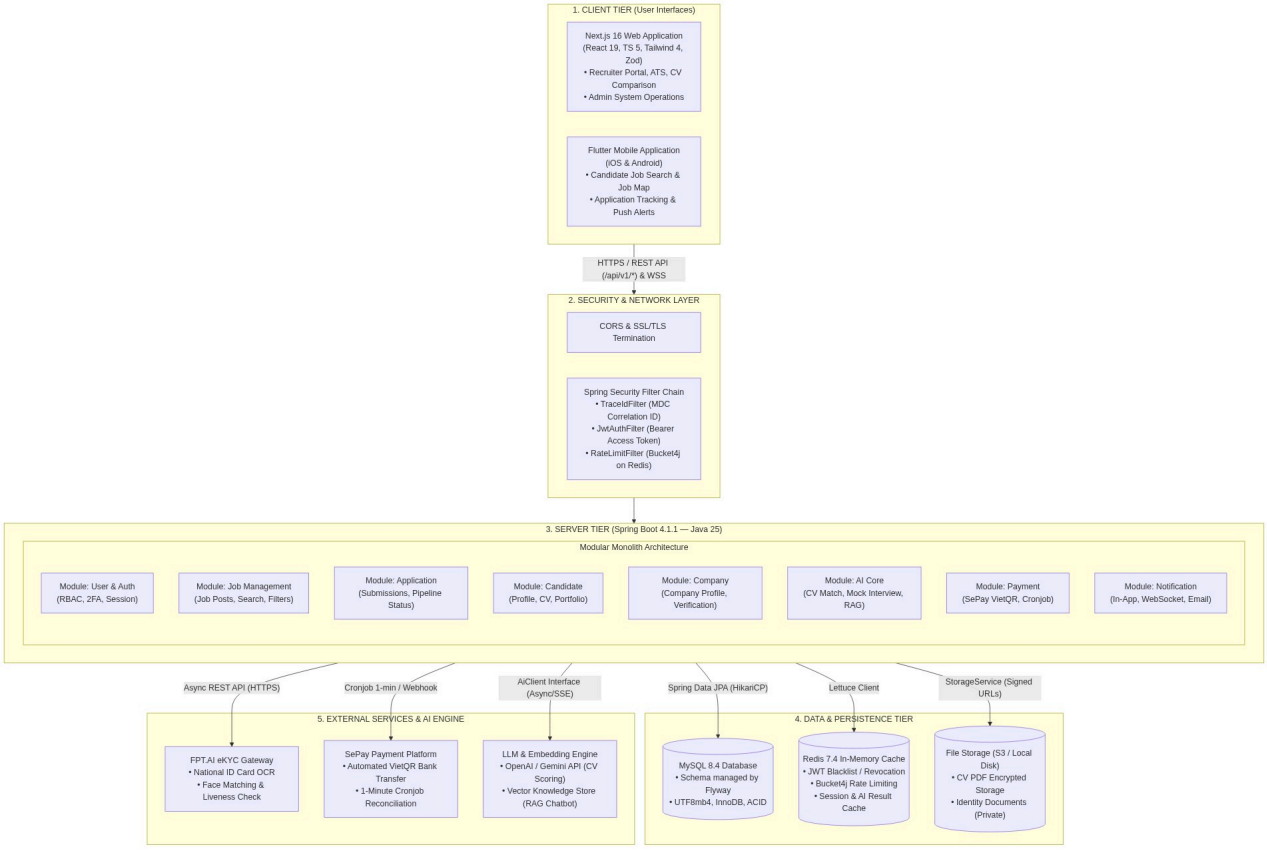


Figure 1.1. Overall Application and Technology Architecture of InterVue Platform

2. Internal Layering & Modular Monolith Pattern

InterVue implements a strictly modular monolith architecture inside Spring Boot 4.1.1. Each domain package enforces clear separation of concerns across controller, service, repository, and event listener components.

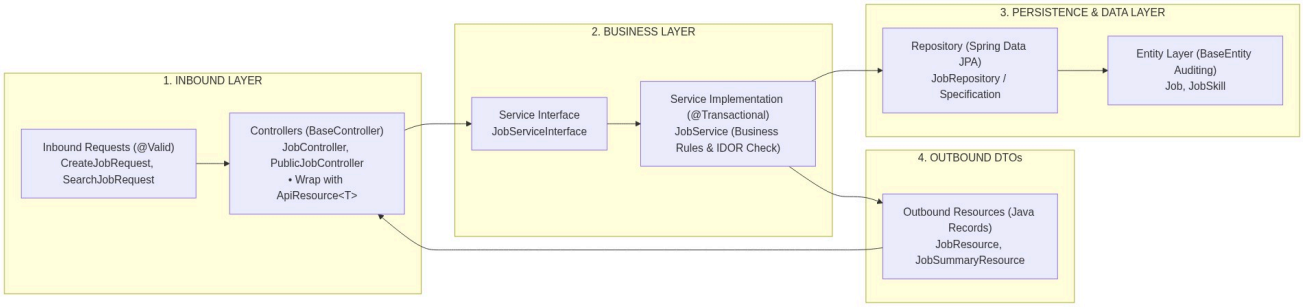


Figure 1.2. Internal Layering and Modular Monolith Pattern

II. Use Cases Diagrams

1. System Use Case Overview

The InterVue system boundary organizes business capabilities across four primary user roles: Guests, Candidates, Recruiters, and Administrators (including Super Admin). The platform coordinates identity verification, recruitment operations, AI-assisted evaluation, and transaction auditing.

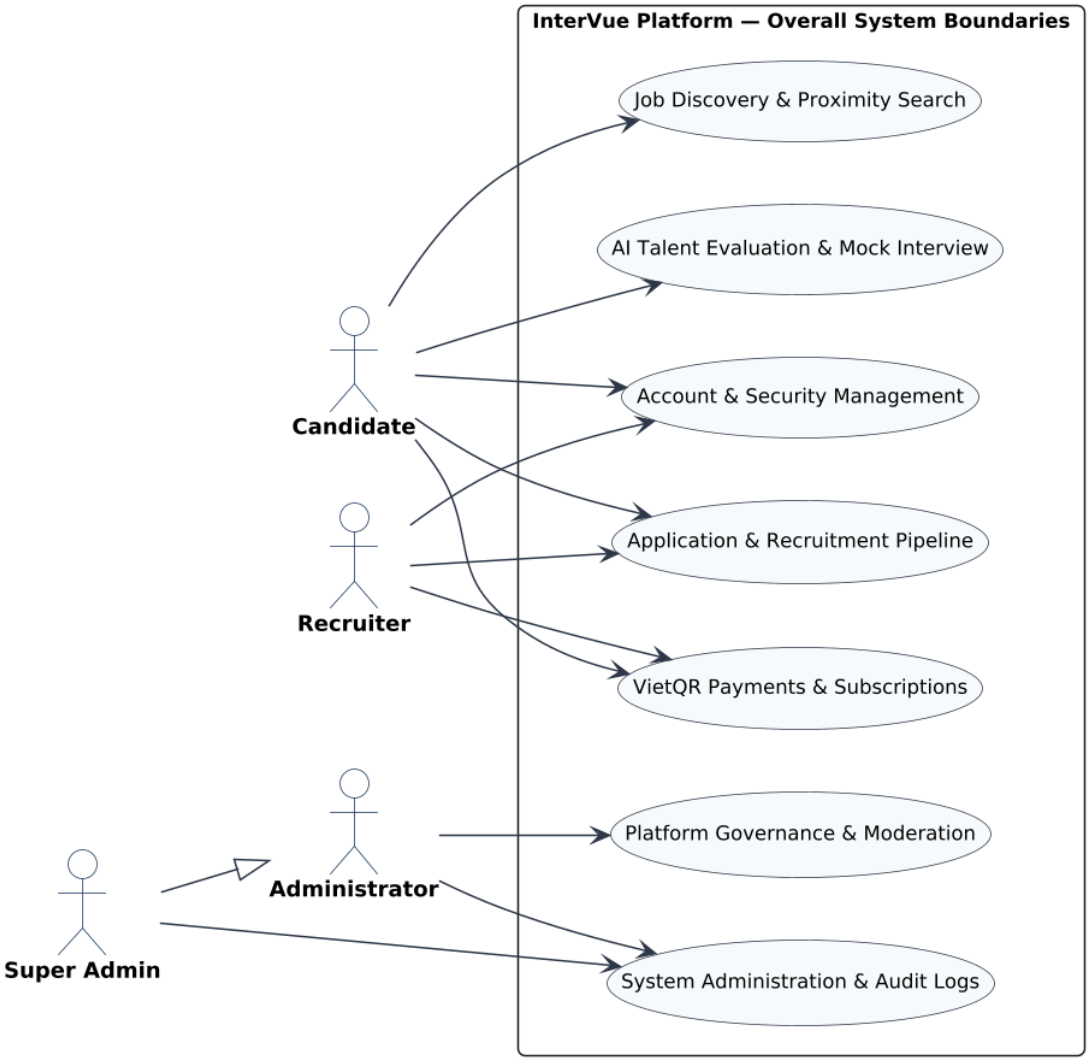


Figure 2.1. Overall System Use Case Overview of InterVue Platform

2. Candidate Subsystem Use Case Model

The Candidate Subsystem manages the end-to-end job seeker journey. Applying for jobs strictly requires verified eKYC status and attachment of a validated CV, with optional extensions for AI-generated cover letters and screening questionnaires.

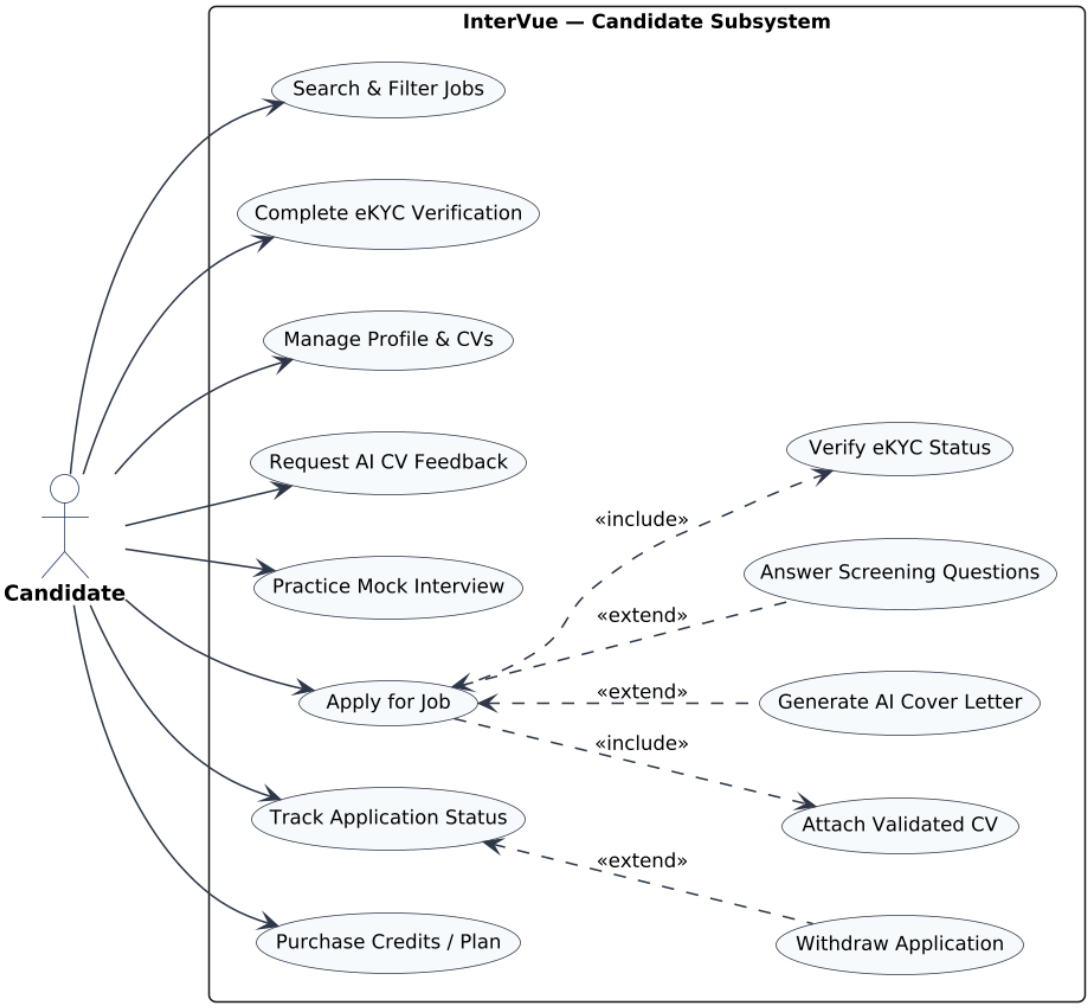


Figure 2.2. Candidate Subsystem Use Case Diagram

3. Recruiter Subsystem Use Case Model

The Recruiter Subsystem empowers talent acquisition teams with company branding, AI-assisted job description authoring, applicant tracking, and automated candidate ranking workflows.

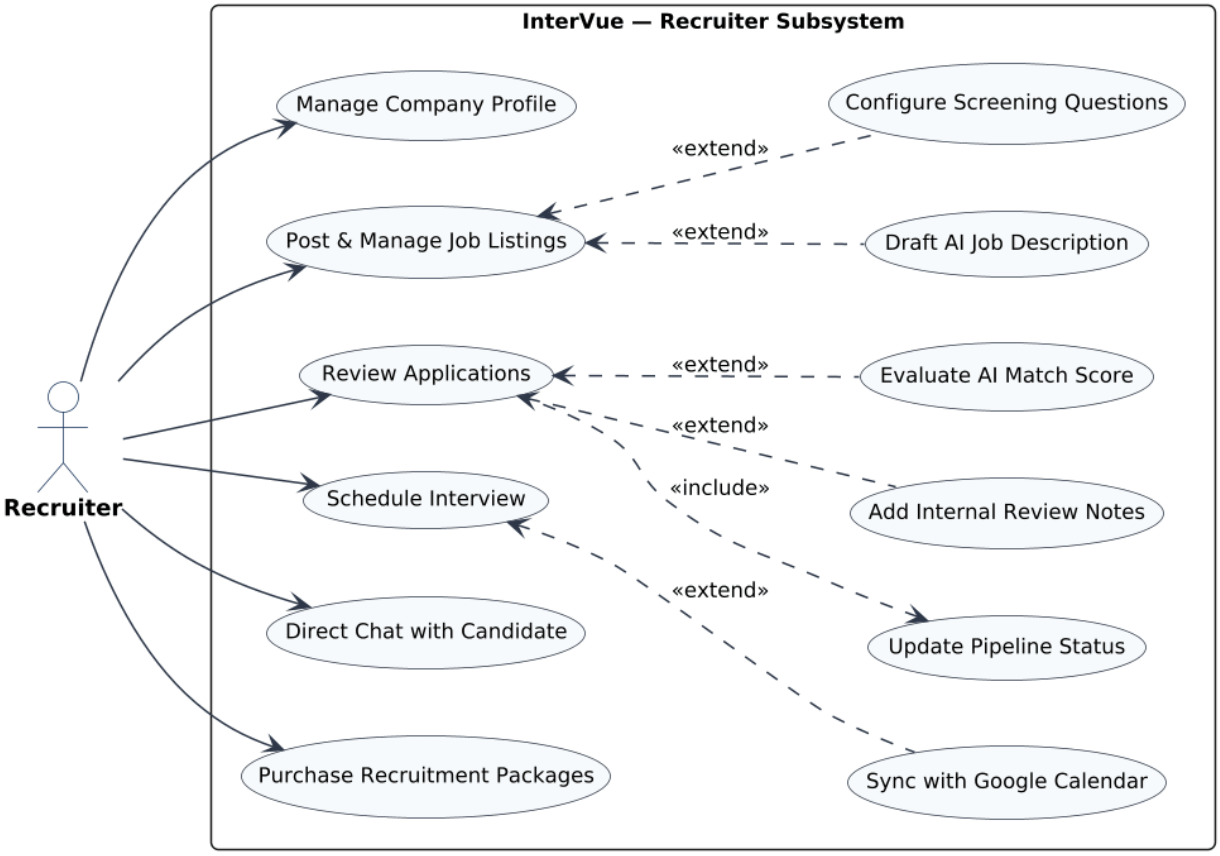


Figure 2.3. Recruiter Subsystem Use Case Diagram

4. Administrator Subsystem Use Case Model

The Administrator Subsystem provides platform operators with governance, moderation, and financial reconciliation tools. The Super Admin actor specializes the Administrator role with exclusive account management authority.

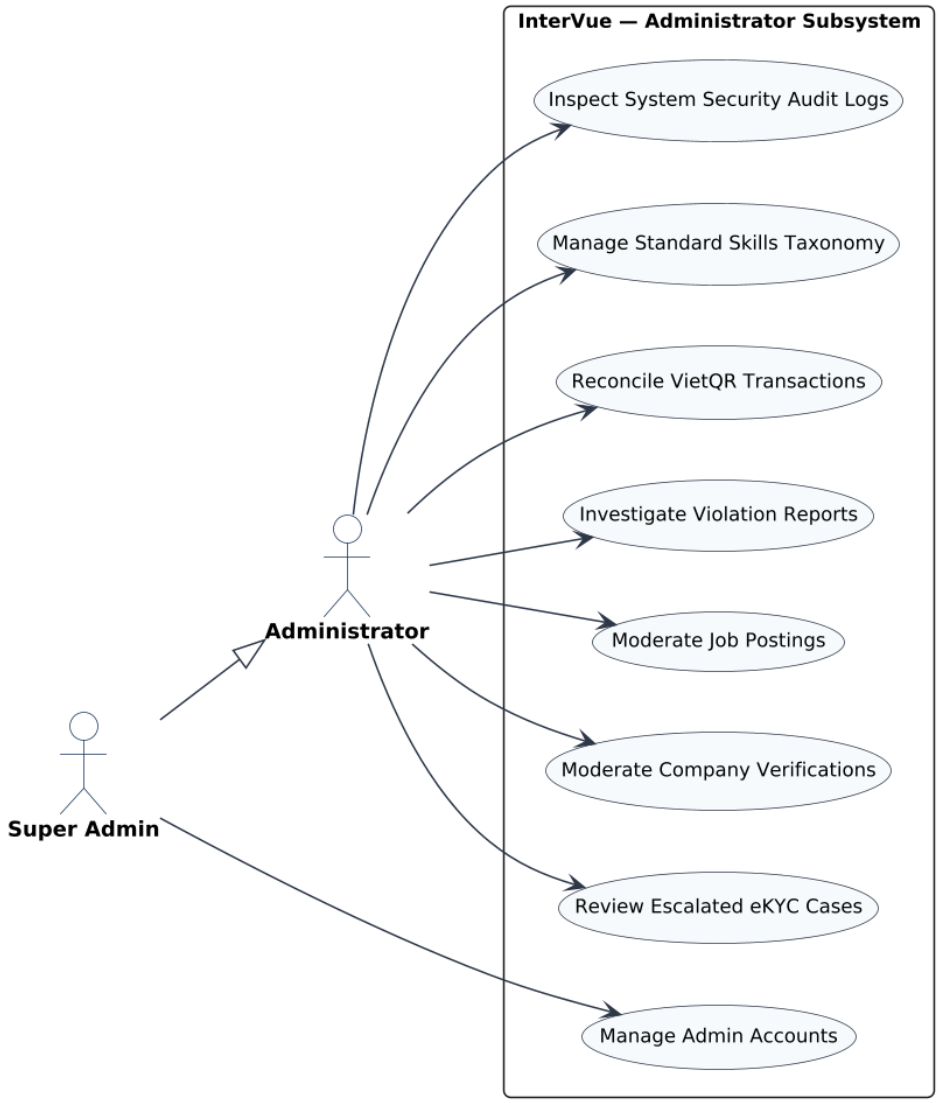


Figure 2.4. Administrator Subsystem Use Case Diagram

III. Sequences Diagrams

1. Authentication & Token Lifecycle Sequence

The authentication sequence diagram depicts the end-to-end token issuance, refresh rotation, and Redis-backed session invalidation lifecycle implemented in InterVue.

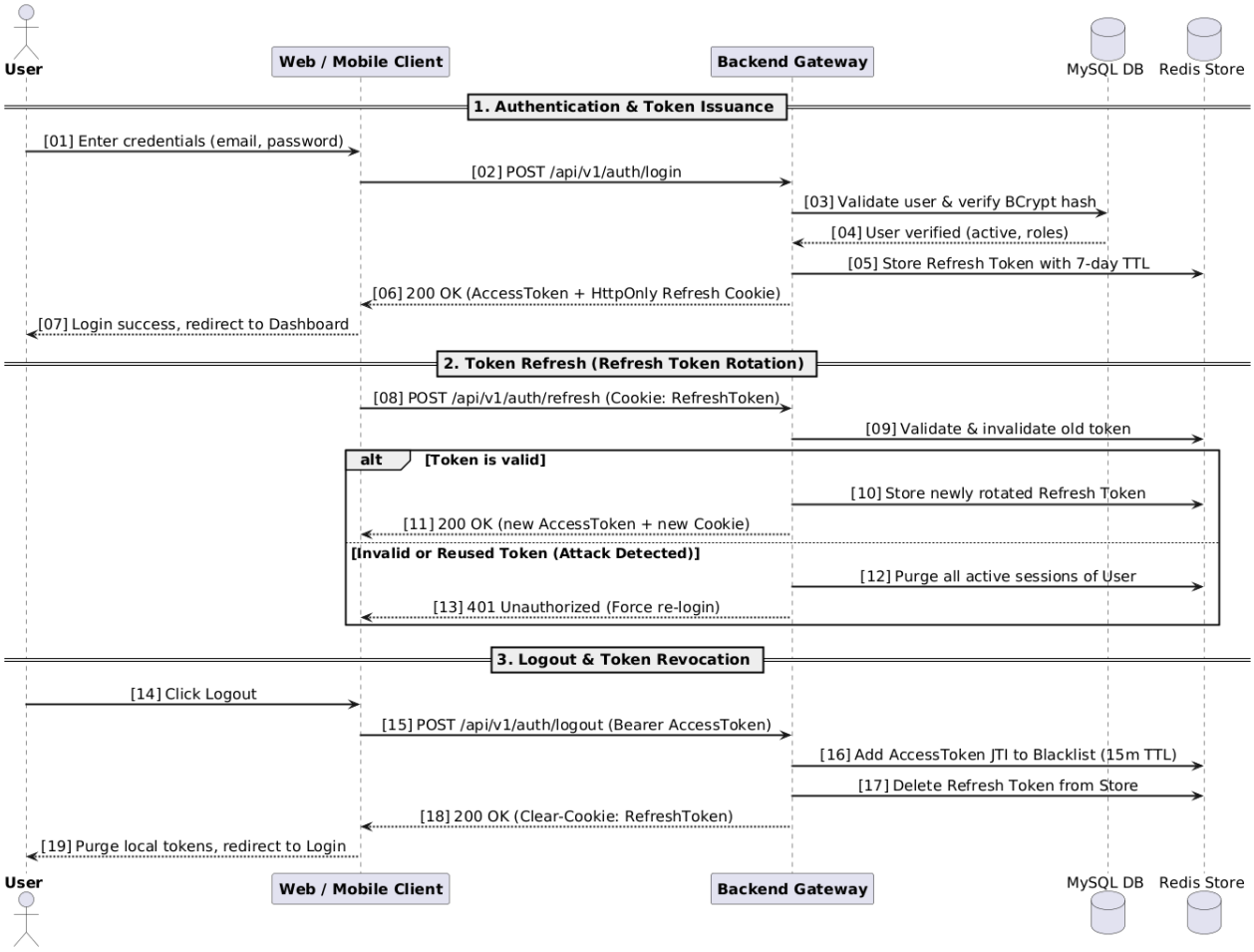


Figure 3.1. Authentication and Token Lifecycle Sequence Diagram

2. Candidate eKYC Identity Verification Workflow

The eKYC verification sequence illustrates automated National ID (CCCD) processing through FPT.AI, facial liveness verification, and administrative fallback escalation for unmatched records.

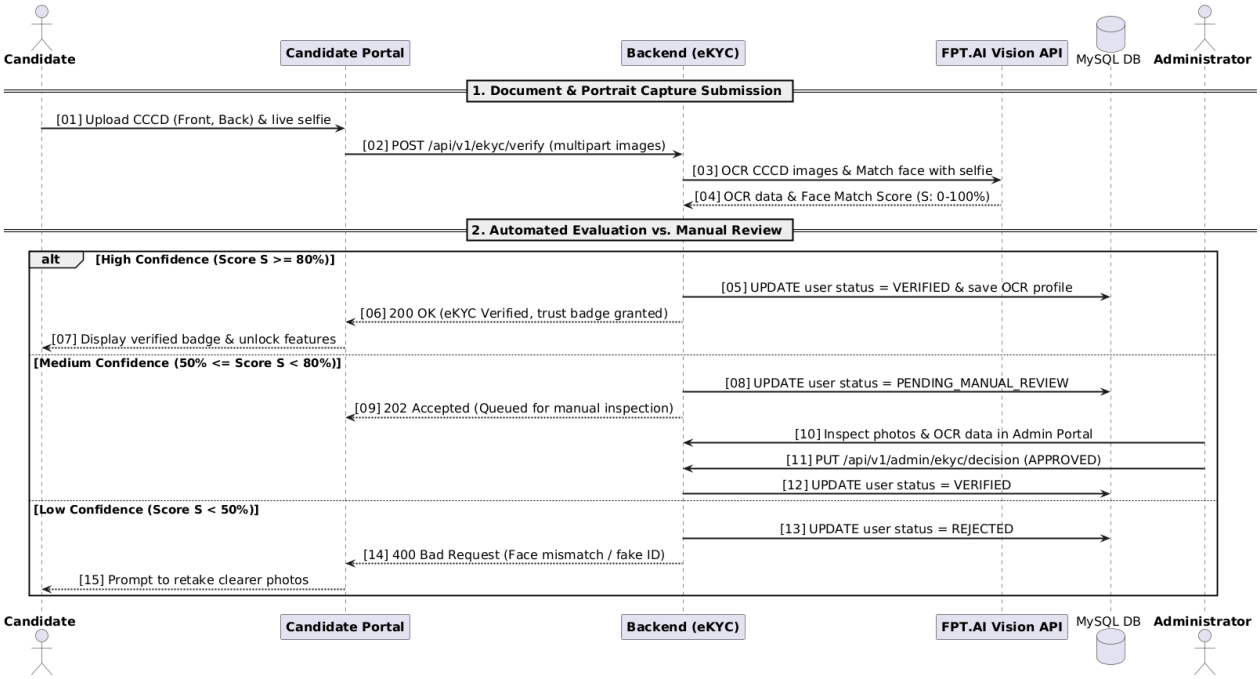


Figure 3.2. Candidate eKYC Identity Verification Sequence Diagram

3. Job Application & Async AI Match Fit Evaluation

The job application sequence illustrates mandatory eKYC gate enforcement, immutable CV snapshotting, and asynchronous AI-assisted candidate fit scoring powered by Gemini LLM.

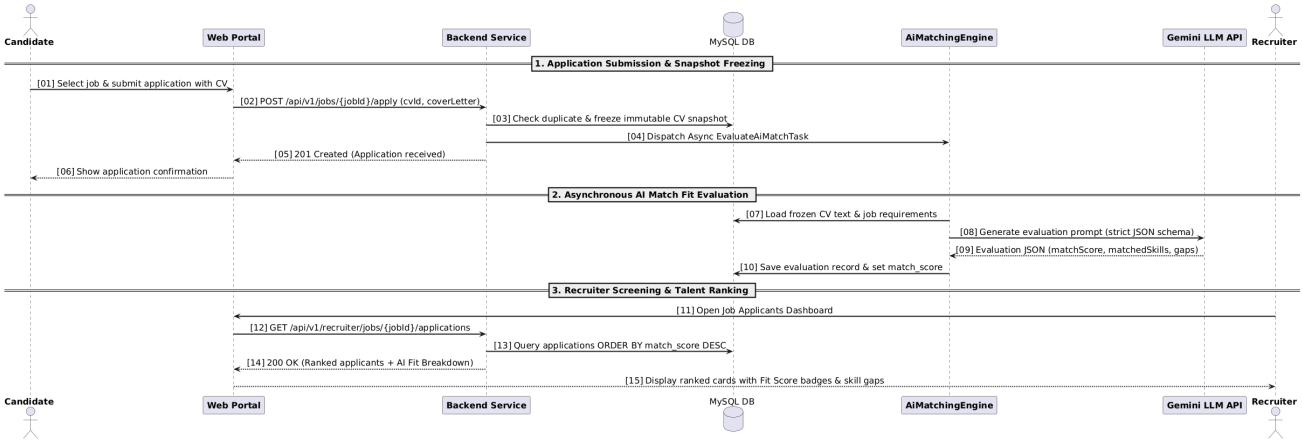


Figure 3.3. Job Application and Asynchronous AI Match Fit Evaluation Sequence Diagram

4. SePay VietQR Payment & 1-Minute Cron Reconciliation

The payment sequence diagram models recruiter service package purchases via Napas 247 dynamic VietQR codes, supported by dual-channel webhook processing and 60-second scheduled cron reconciliation.

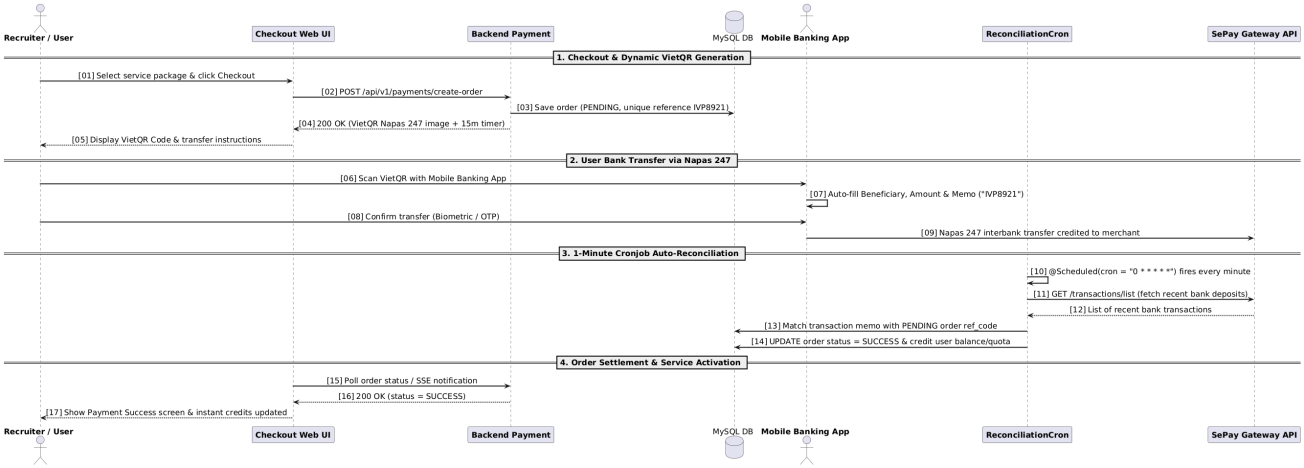


Figure 3.4. SePay VietQR Payment and Scheduled Cron Reconciliation Sequence Diagram

IV. Class Diagrams

1. Identity, Recruitment & Job Catalog Domain Model

The domain class diagram below models the core recruitment catalog and identity entities, illustrating relationships among user accounts, candidate profiles, verified company entities, job postings, and skill taxonomies.

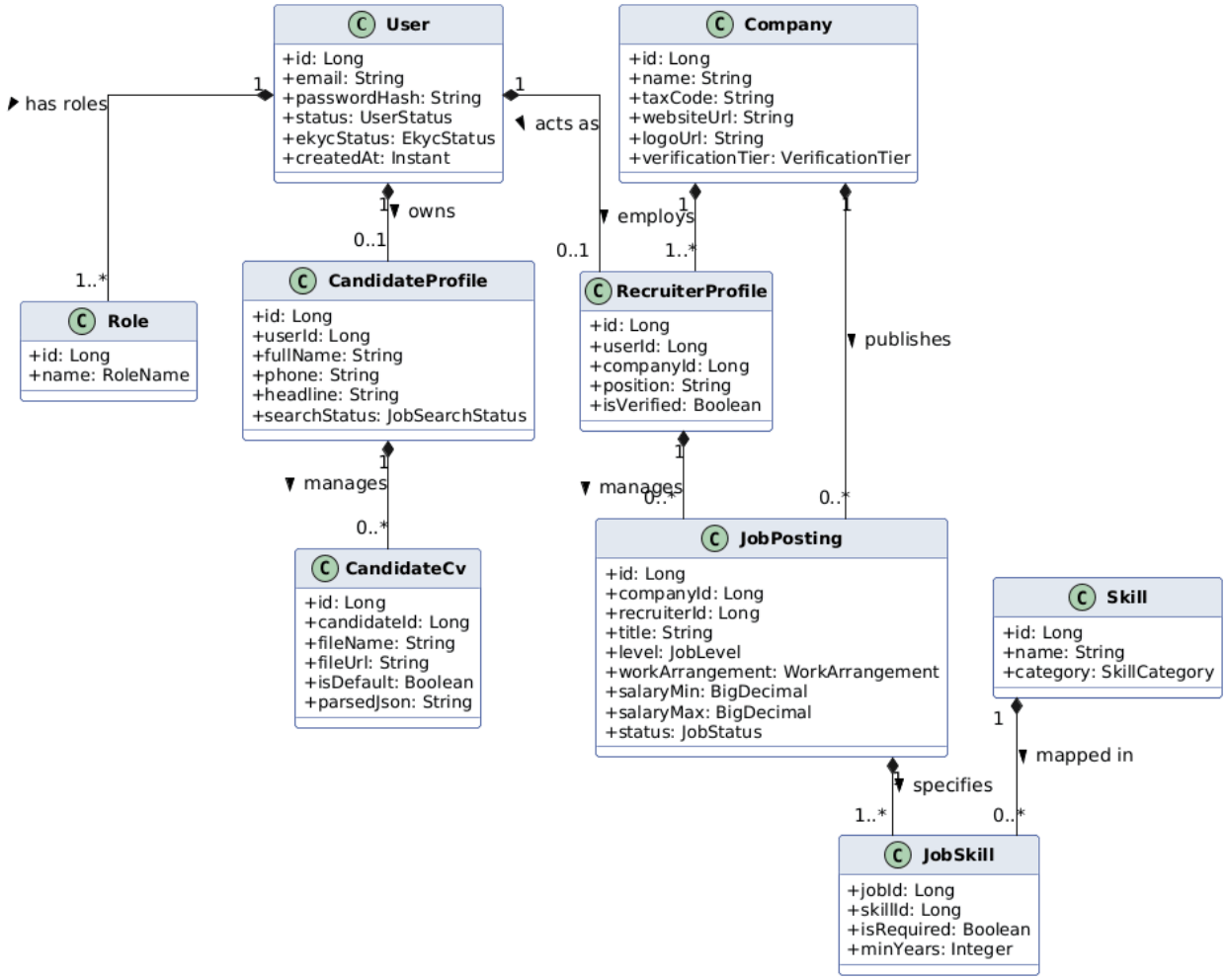


Figure 4.1. Identity, Recruitment and Job Catalog Class Diagram

2. Application Pipeline, AI Assessment & Financial Settlement Domain Model

The domain class diagram below illustrates the operational lifecycle of candidate applications, automated AI match evaluation, interactive mock interview turns, and VietQR financial settlement.

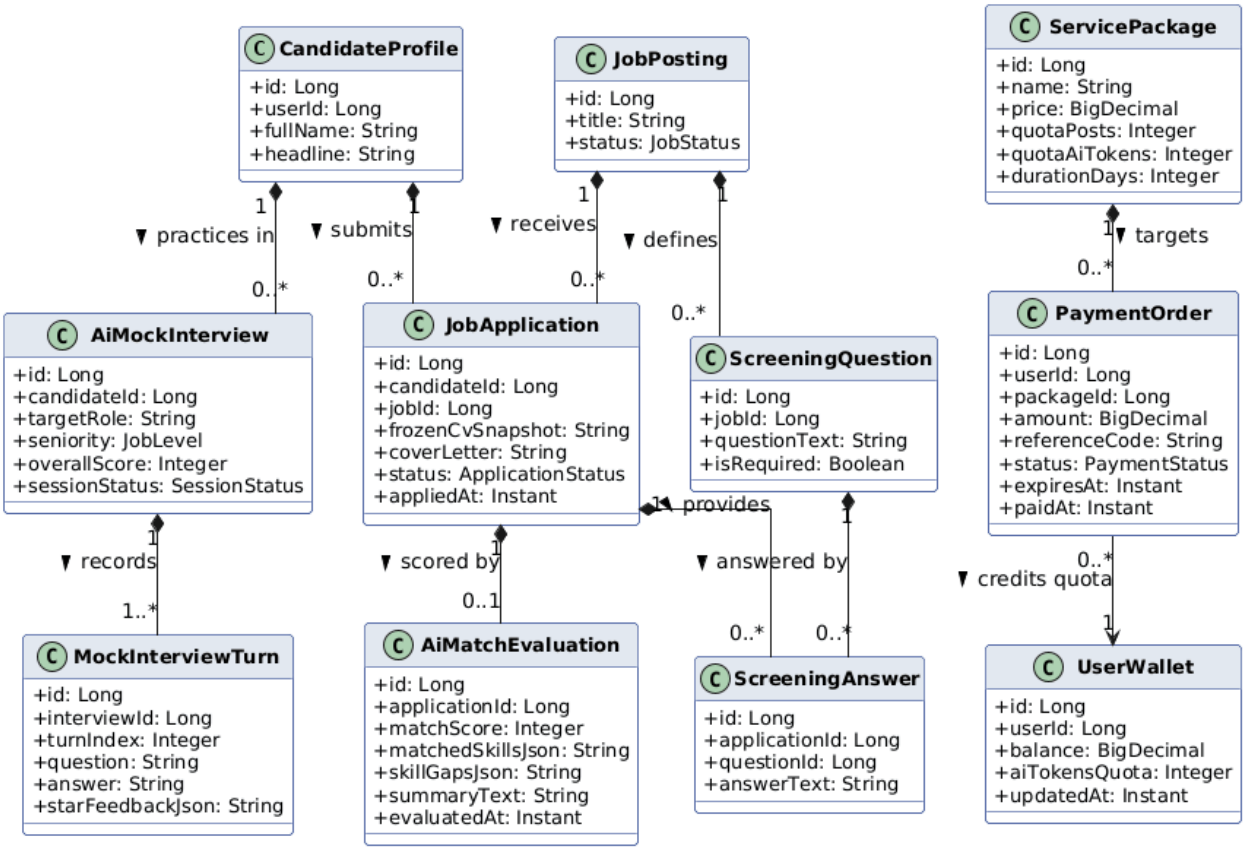


Figure 4.2. Application Pipeline, AI Assessment and Financial Settlement Class Diagram

V. Data Flow Diagrams (DFD)

1. System Context Diagram (DFD Level 0)

The Level 0 Data Flow Diagram (Context Diagram) establishes the overarching boundary of the InterVue ecosystem, mapping information flows between primary human user roles, the centralized platform processing core, and third-party external service gateways.

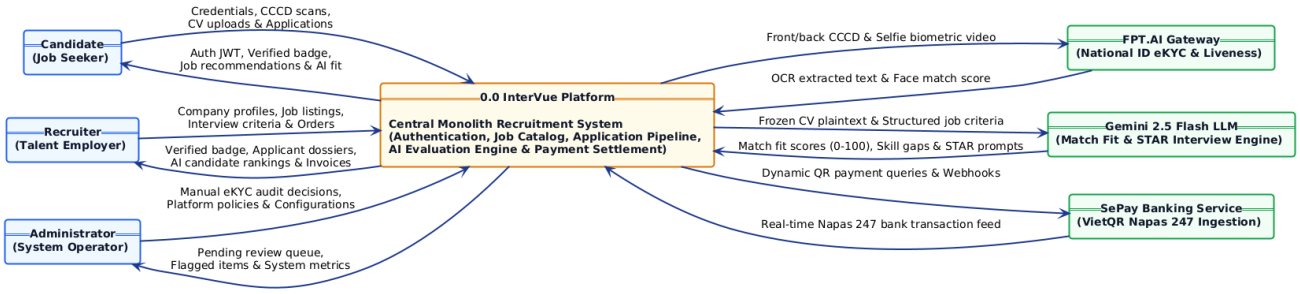


Figure 5.1. System Context Diagram (DFD Level 0) of InterVue Platform

2. Subsystem Process Decomposition (DFD Level 1)

The Level 1 Data Flow Diagram decomposes the InterVue platform into four core functional subsystems, mapping transactional data exchanges between user roles, external service gateways, and four dedicated persistent data stores.

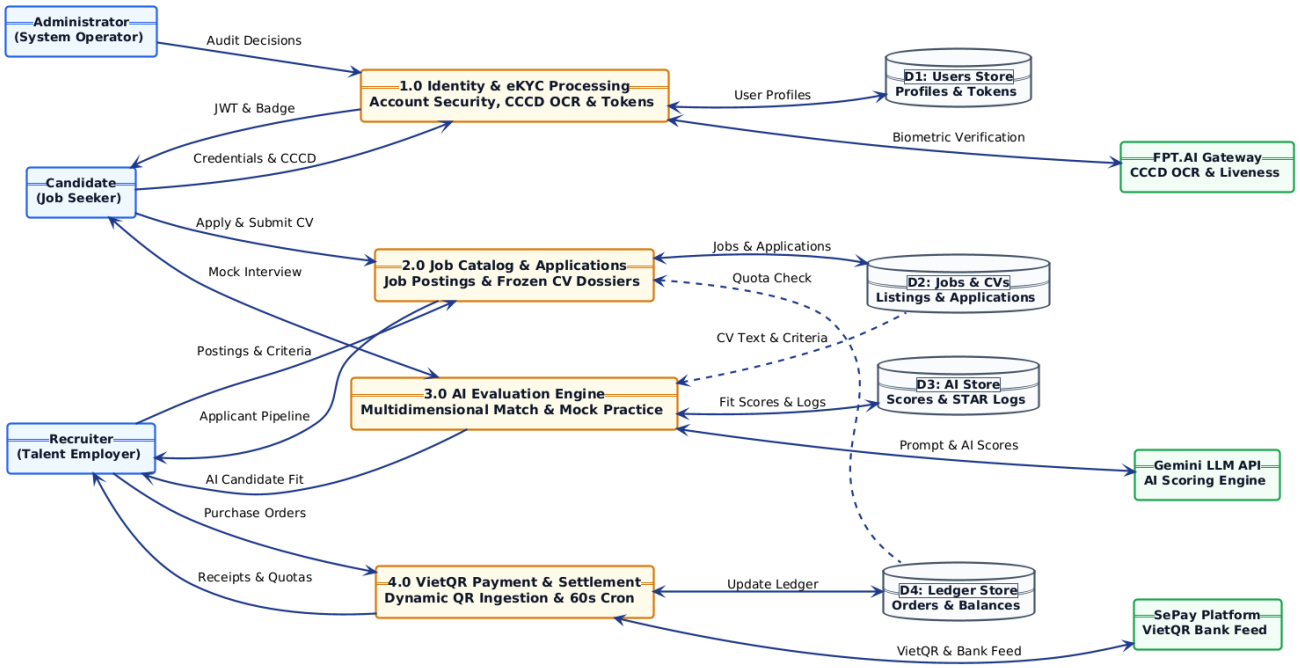


Figure 5.2. Subsystem Process Decomposition Diagram (DFD Level 1)

VI. Database Design

1. Entity Relationship Diagram

The Entity Relationship Diagram provides a comprehensive conceptual and physical relational model of the InterVue platform, mapping all sixteen persistent data entities across identity management, candidate dossiers, corporate recruitment pipelines, AI evaluation engines, and VietQR financial settlement.

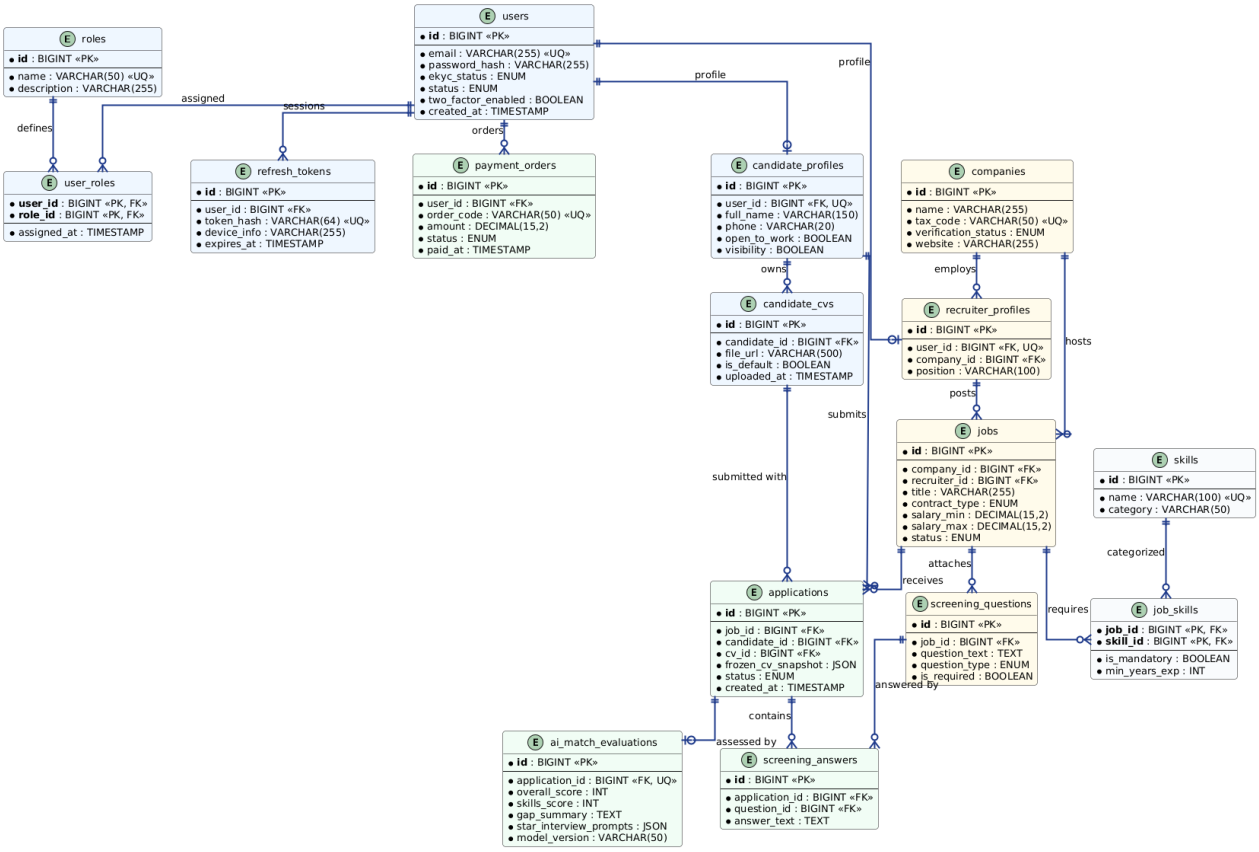


Figure 6.1. Comprehensive Entity Relationship Diagram (ERD) of InterVue Platform

2. Schema Overview

The schema overview categorizes all sixteen relational database tables implemented on MySQL 8.4 utilizing InnoDB storage engines and utf8mb4 character collation, outlining the primary functional scope and storage responsibility of each relational entity.

STT	Name	Description
1	**roles**	Stores master system roles for Spring Security RBAC authorization (CANDIDATE, RECRUITER, ADMIN)
2	**users**	Manages central user authentication accounts, eKYC status, 2FA settings, and soft deletion
3	**user_roles**	Associative join entity enabling many-to-many role assignments per user account
4	**refresh_tokens**	Tracks opaque hashed JWT refresh tokens, device fingerprints, and session expiry
5	**candidate_profiles**	Holds candidate profile info, contact details, career headline, and job search visibility
6	**candidate_cvs**	Manages uploaded CV document files, cloud storage URLs, and default resume flags
7	**companies**	Stores corporate employer entities, tax identification codes, and verification status
8	**recruiter_profiles**	Associates enterprise recruitment staff accounts with verified corporate employers
9	**jobs**	Manages job vacancy postings, contract types, work arrangements, salary bands, and lifecycle
10	**skills**	Master catalog of standardized IT technical skills and competencies
11	**job_skills**	Maps mandatory and optional technical skill requirements to job vacancies
12	**screening_questions**	Custom screening questions attached to jobs (Text, Single Choice, Multiple Choice)
13	**applications**	Candidate job application submissions containing frozen CV snapshots and pipeline states
14	**screening_answers**	Applicant responses to employer screening questions attached to applications
15	**ai_match_evaluations**	AI-generated compatibility scores (0–100), skill GAP analysis, and STAR interview prompts
16	**payment_orders**	Financial transaction ledger for VietQR recruitment service package purchases

Table 6.1. Relational Database Schema Overview of InterVue Platform

3. Detail Schema

The detailed schema specification defines the complete physical field catalog, data types, constraints, and operational logic for all relational database entities.

3.1. roles

roles						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Primary key surrogate identifier of system role
2		No	**name**	VARCHAR(50)		Unique role name identifier (CANDIDATE, RECRUITER, ADMIN)
3		Yes	**description**	VARCHAR(255)		Human-readable description of role permissions

3.2. users

users						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Unique account identifier
2		No	**email**	VARCHAR(255)		Unique user email address used for login and notification
3		No	**password_hash**	VARCHAR(255)		Salted BCrypt cryptographic hash of user password
4		No	**status**	ENUM		Account state: ACTIVE, INACTIVE, BANNED
5		No	**ekyc_status**	ENUM		Identity state: NOT_SUBMITTED, PENDING, APPROVED, REJECTED
6		No	**is_email_verified**	BOOLEAN		Email verification token confirmation flag
7		No	**two_factor_enabled**	BOOLEAN		TOTP two-factor authentication activation toggle
8		No	**created_at**	TIMESTAMP		UTC record creation timestamp
9		No	**updated_at**	TIMESTAMP		UTC record last update timestamp
10		Yes	**deleted_at**	TIMESTAMP		UTC soft-deletion timestamp for reversible account removal

3.3. user_roles

user_roles						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK, FK**	No	**user_id**	BIGINT	users(id)	Reference to user account holding assigned role
2	**PK, FK**	No	**role_id**	BIGINT	roles(id)	Reference to granted system authority role
3		No	**assigned_at**	TIMESTAMP		UTC timestamp when role was granted to account

3.4. refresh_tokens

refresh_tokens						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Primary surrogate key of session record
2	**FK**	No	**user_id**	BIGINT	users(id)	Reference to authenticated user account
3		No	**token_hash**	VARCHAR(64)		SHA-256 hash of opaque refresh token cookie
4		Yes	**device_info**	VARCHAR(255)		Client browser agent and IP telemetry metadata
5		No	**expires_at**	TIMESTAMP		Absolute expiration timestamp of token family
6		Yes	**revoked_at**	TIMESTAMP		Revocation timestamp if manually logged out or rotated
7		No	**created_at**	TIMESTAMP		UTC session establishment timestamp

3.5. candidate_profiles

candidate_profiles						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Primary key of candidate profile record
2	**FK**	No	**user_id**	BIGINT	users(id)	One-to-one foreign key reference to user account
3		No	**full_name**	VARCHAR(150)		Full legal candidate name verified via eKYC CCCD
4		Yes	**phone**	VARCHAR(20)		Candidate contact telephone number
5		Yes	**headline**	VARCHAR(255)		Professional career summary or headline title
6		Yes	**bio**	TEXT		Detailed autobiographical background and intro
7		No	**open_to_work**	BOOLEAN		Flag indicating candidate is actively seeking employment
8		No	**visibility**	BOOLEAN		Privacy toggle permitting profile discovery by recruiters
9		No	**created_at**	TIMESTAMP		UTC profile creation timestamp
10		No	**updated_at**	TIMESTAMP		UTC profile last modification timestamp

3.6. candidate_cvs

candidate_cvs						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Unique CV asset identifier
2	**FK**	No	**candidate_id**	BIGINT	candidate_profiles(id)	Reference to owner candidate profile
3		No	**file_url**	VARCHAR(500)		Secure cloud storage object key or presigned URL
4		No	**file_name**	VARCHAR(255)		Original file name uploaded by candidate
5		No	**is_default**	BOOLEAN		Flag designating CV as primary default application resume
6		No	**uploaded_at**	TIMESTAMP		UTC upload timestamp

3.7. companies

companies						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Primary corporate employer identifier
2		No	**name**	VARCHAR(255)		Official registered legal name of enterprise
3		No	**tax_code**	VARCHAR(50)		Unique national corporate tax identification code
4		Yes	**website**	VARCHAR(255)		Official enterprise corporate web URL
5		Yes	**logo_url**	VARCHAR(500)		Cloud storage key for enterprise logo branding image
6		No	**verification_status**	ENUM		Audit state: UNVERIFIED, PENDING, VERIFIED, REJECTED
7		No	**created_at**	TIMESTAMP		UTC enterprise registration timestamp
8		No	**updated_at**	TIMESTAMP		UTC enterprise verification update timestamp

3.8. recruiter_profiles

recruiter_profiles						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Unique recruiter personnel profile identifier
2	**FK**	No	**user_id**	BIGINT	users(id)	One-to-one reference to recruiter user account
3	**FK**	No	**company_id**	BIGINT	companies(id)	Reference to corporate employer represented by recruiter
4		Yes	**position**	VARCHAR(100)		Corporate job title (e.g., HR Manager, Tech Recruiter)
5		No	**created_at**	TIMESTAMP		UTC recruiter profile creation timestamp
6		No	**updated_at**	TIMESTAMP		UTC recruiter profile last update timestamp

3.9. jobs

jobs

No.	Key	Nullable	Field name	Type	Reference Table	Description
1	**PK**	No	**id**	BIGINT		Primary key vacancy posting identifier
2	**FK**	No	**company_id**	BIGINT	companies(id)	Associated verified corporate employer
3	**FK**	No	**recruiter_id**	BIGINT	recruiter_profiles(id)	Recruiter author who published the vacancy
4		No	**title**	VARCHAR(255)		Job opening headline title
5		No	**contract_type**	ENUM		Contract type: FULL_TIME, PART_TIME, INTERNSHIP
6		No	**working_arrangement**	ENUM		Work mode: ONSITE, HYBRID, REMOTE, FLEXIBLE
7		Yes	**salary_min**	DECIMAL(15,2)		Minimum monthly compensation limit in VND
8		Yes	**salary_max**	DECIMAL(15,2)		Maximum monthly compensation limit in VND
9		No	**currency**	VARCHAR(10)		Currency code (Default: VND)
10		No	**status**	ENUM		Lifecycle status: DRAFT, PUBLISHED, CLOSED
11		Yes	**published_at**	TIMESTAMP		Official vacancy publication timestamp
12		No	**created_at**	TIMESTAMP		Record creation timestamp
13		No	**updated_at**	TIMESTAMP		Last modification timestamp

3.10. skills

skills

No.	Key	Nullable	Field name	Type	Reference Table	Description
1	**PK**	No	**id**	BIGINT		Unique skill identifier in taxonomy
2		No	**name**	VARCHAR(100)		Standardized technical skill name (e.g., Java, React, Docker)
3		Yes	**category**	VARCHAR(50)		Technical classification: BACKEND, FRONTEND, DEVOPS, AI
4		No	**created_at**	TIMESTAMP		UTC taxonomy entry creation timestamp

3.11. job_skills

job_skills						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK, FK**	No	**job_id**	BIGINT	jobs(id)	Reference to target job posting
2	**PK, FK**	No	**skill_id**	BIGINT	skills(id)	Reference to standardized taxonomy skill
3		No	**is_mandatory**	BOOLEAN		Flag designating skill as strictly non-negotiable prerequisite
4		Yes	**min_years_exp**	INT		Minimum required years of commercial experience

3.12. screening_questions

screening_questions						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Unique screening question identifier
2	**FK**	No	**job_id**	BIGINT	jobs(id)	Reference to parent job posting
3		No	**question_text**	TEXT		Screening questionnaire prompt presented to applicant
4		No	**question_type**	ENUM		Input type: TEXT, SINGLE_CHOICE, MULTI_CHOICE
5		No	**is_required**	BOOLEAN		Mandatory answer flag before application submission

3.13. applications

applications						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Primary application transaction identifier
2	**FK**	No	**job_id**	BIGINT	jobs(id)	Target job vacancy opening
3	**FK**	No	**candidate_id**	BIGINT	candidate_profiles(id)	Applying candidate profile
4	**FK**	No	**cv_id**	BIGINT	candidate_cvs(id)	Selected candidate CV asset
5		No	**frozen_cv_snapshot**	JSON		Immutable JSON snapshot of CV attributes captured at apply time
6		No	**status**	ENUM		Pipeline state: APPLIED, IN_REVIEW, INTERVIEW, OFFER, HIRED
7		Yes	**cover_letter**	TEXT		Personal cover letter or AI-assisted pitch statement
8		No	**created_at**	TIMESTAMP		Application submission timestamp
9		No	**updated_at**	TIMESTAMP		Application pipeline stage transition timestamp

3.14. screening_answers

screening_answers						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Unique screening answer record identifier
2	**FK**	No	**application_id**	BIGINT	applications(id)	Associated candidate job application
3	**FK**	No	**question_id**	BIGINT	screening_questions(id)	Corresponding screening question answered
4		No	**answer_text**	TEXT		Candidate response content or selected option values
5		No	**created_at**	TIMESTAMP		UTC answer submission timestamp

3.15. ai_match_evaluations

ai_match_evaluations						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		AI evaluation result record identifier
2	**FK**	No	**application_id**	BIGINT	applications(id)	One-to-one unique link to evaluated job application
3		No	**overall_score**	INT		Overall compatibility rating (0-100) by Gemini LLM
4		No	**skills_score**	INT		Technical skill alignment score (0-100)
5		No	**experience_score**	INT		Seniority and work experience alignment score (0-100)
6		Yes	**strength_summary**	TEXT		Bullet summary of candidate strengths against job posting
7		Yes	**gap_summary**	TEXT		Detailed gap analysis of missing or unmentioned qualifications
8		Yes	**star_interview_prompts**	JSON		Tailored behavioral interview questions targeting GAP areas
9		No	**model_version**	VARCHAR(50)		Active Gemini LLM model identifier for ethical auditability
10		No	**evaluated_at**	TIMESTAMP		UTC evaluation completion timestamp

3.16. payment_orders

payment_orders						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Financial payment order identifier
2	**FK**	No	**user_id**	BIGINT	users(id)	Purchasing employer or candidate account
3		No	**order_code**	VARCHAR(50)		Unique VietQR transfer reference code (e.g., INV-202610-8832)
4		No	**amount**	DECIMAL(15,2)		Monetary settlement amount in VND currency
5		No	**payment_method**	VARCHAR(50)		Payment gateway method (Default: VIETQR)
6		No	**status**	ENUM		Billing status: PENDING, PAID, EXPIRED, FAILED
7		Yes	**sepay_transaction_id**	VARCHAR(100)		Inbound bank transfer ID reconciled via SePay webhook
8		Yes	**paid_at**	TIMESTAMP		UTC financial confirmation timestamp
9		No	**created_at**	TIMESTAMP		Order initiation timestamp

VII. SiteMap

1. Web SiteMap by Route Group

The Web SiteMap illustrates the hierarchical route organization of the InterVue web portal implemented with Next.js 16 App Router, segregating public traffic, identity flows, candidate workspaces, enterprise recruiter portals, and administrative governance modules.

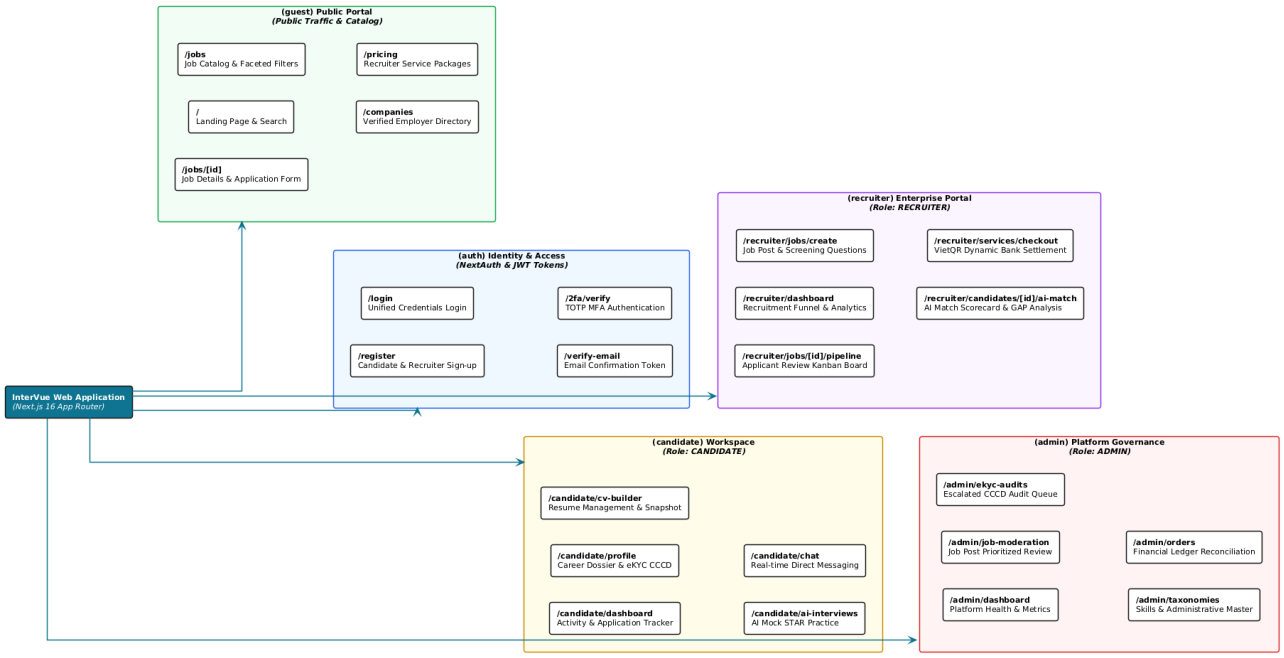


Figure 7.1. Web SiteMap & Route Group Architecture of InterVue Platform

2. Mobile SiteMap & Screen Navigation Hierarchy

The Mobile SiteMap maps the screen navigation hierarchy and stateful widget trees of the InterVue Flutter mobile application, providing candidates with seamless native workflows for job discovery, biometric authentication, application tracking, and recruiter chat.

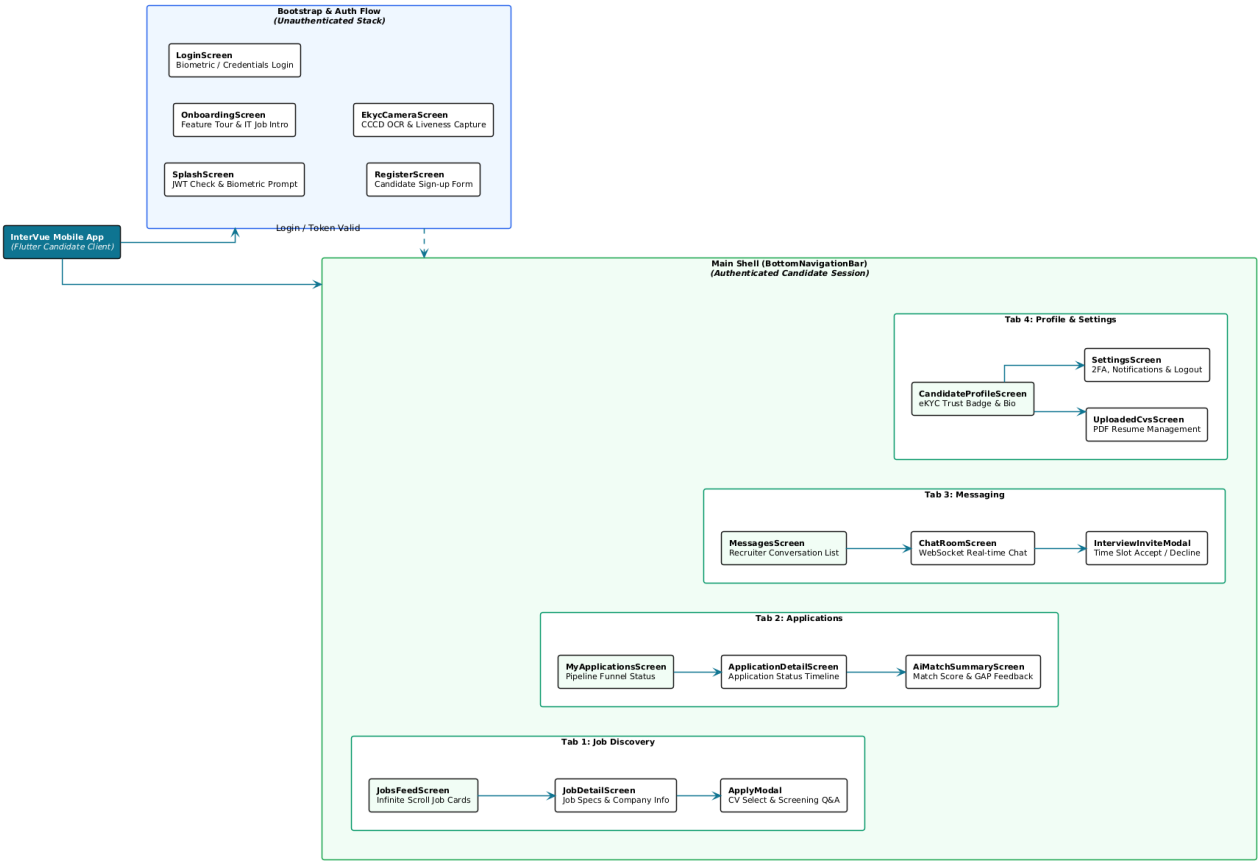


Figure 7.2. Mobile Screen Navigation Hierarchy & Widget Stack of InterVue Candidate App

VIII. Core Algorithms

1. AI Match Fit Evaluation & Scoring Pipeline

The AI Match Fit Evaluation algorithm orchestrates the ingestion of immutable application snapshots, automated prompt synthesis with schema constraints, and asynchronous evaluation via Google Gemini 1.5 Flash to generate explainable candidate compatibility scores.

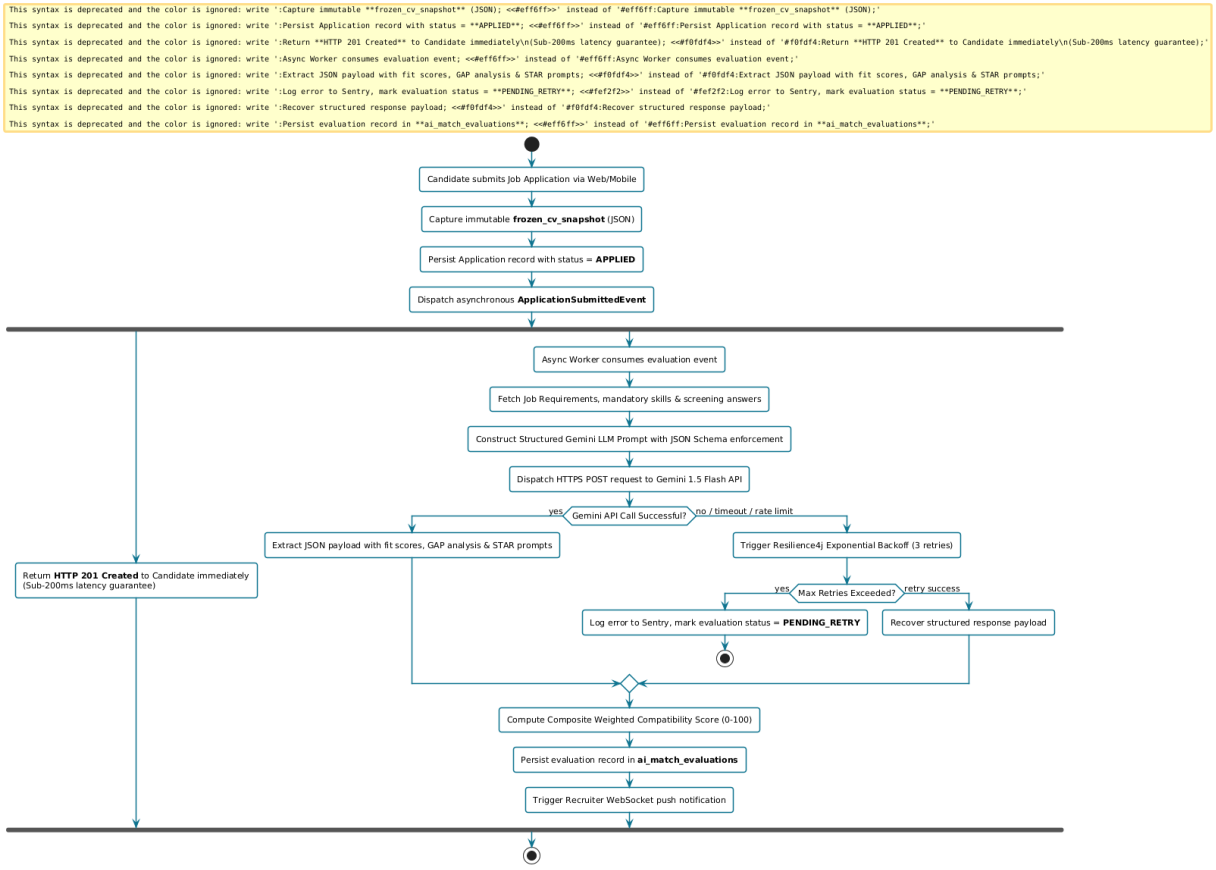


Figure 8.1. Asynchronous AI Match Fit Evaluation & Scoring Pipeline Flowchart

2. Candidate eKYC Identity Verification Workflow

The eKYC identity verification algorithm governs optical character recognition, anti-spoofing fraud detection, and facial biometric matching between national citizen ID cards (CCCD) and live candidate portrait selfies via FPT.AI API integration.

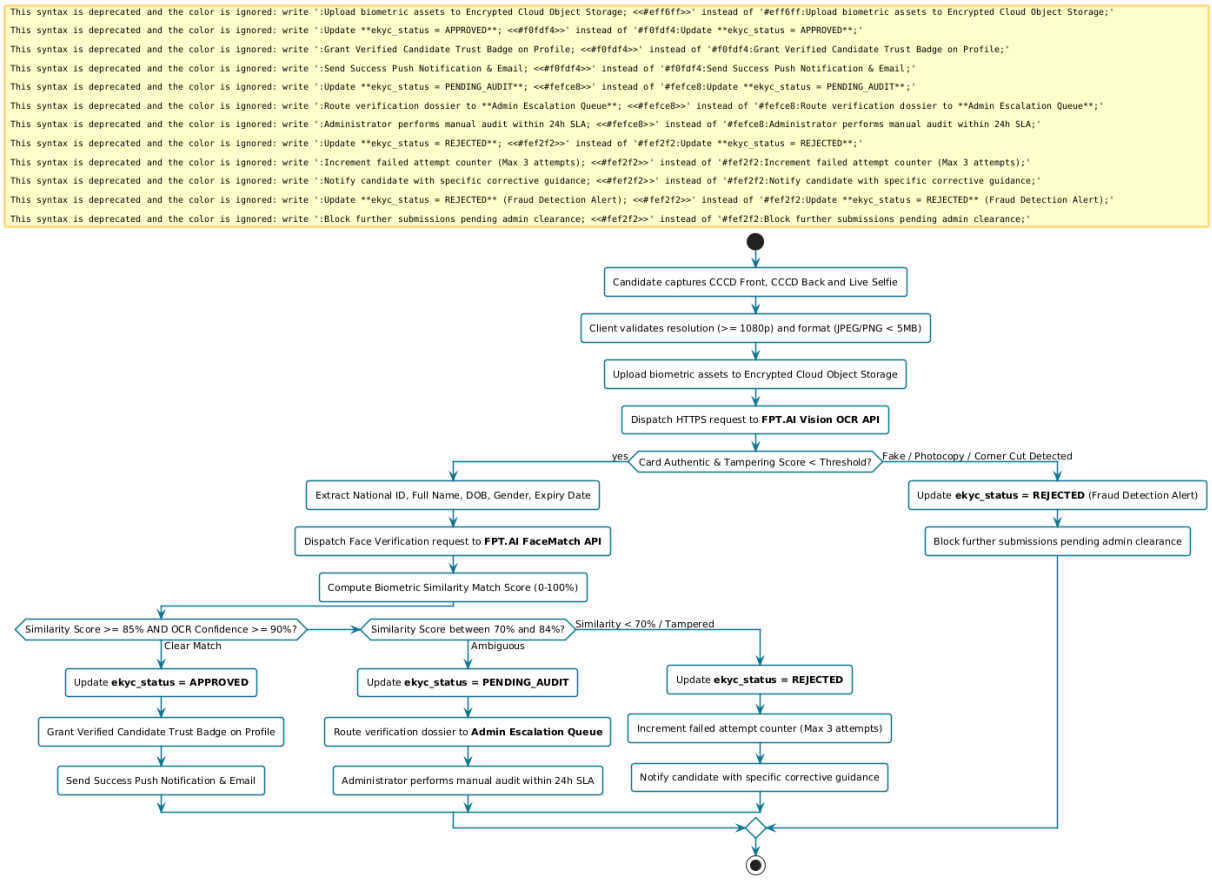


Figure 8.2. Multi-Stage eKYC Identity Verification & Anti-Spoofing Workflow Flowchart

3. Idempotent VietQR Payment & Reconciliation Processing

The VietQR payment algorithm guarantees atomic financial settlement, idempotent webhook processing, and dual-track reconciliation across inbound SePay banking webhooks and automated background polling workers.

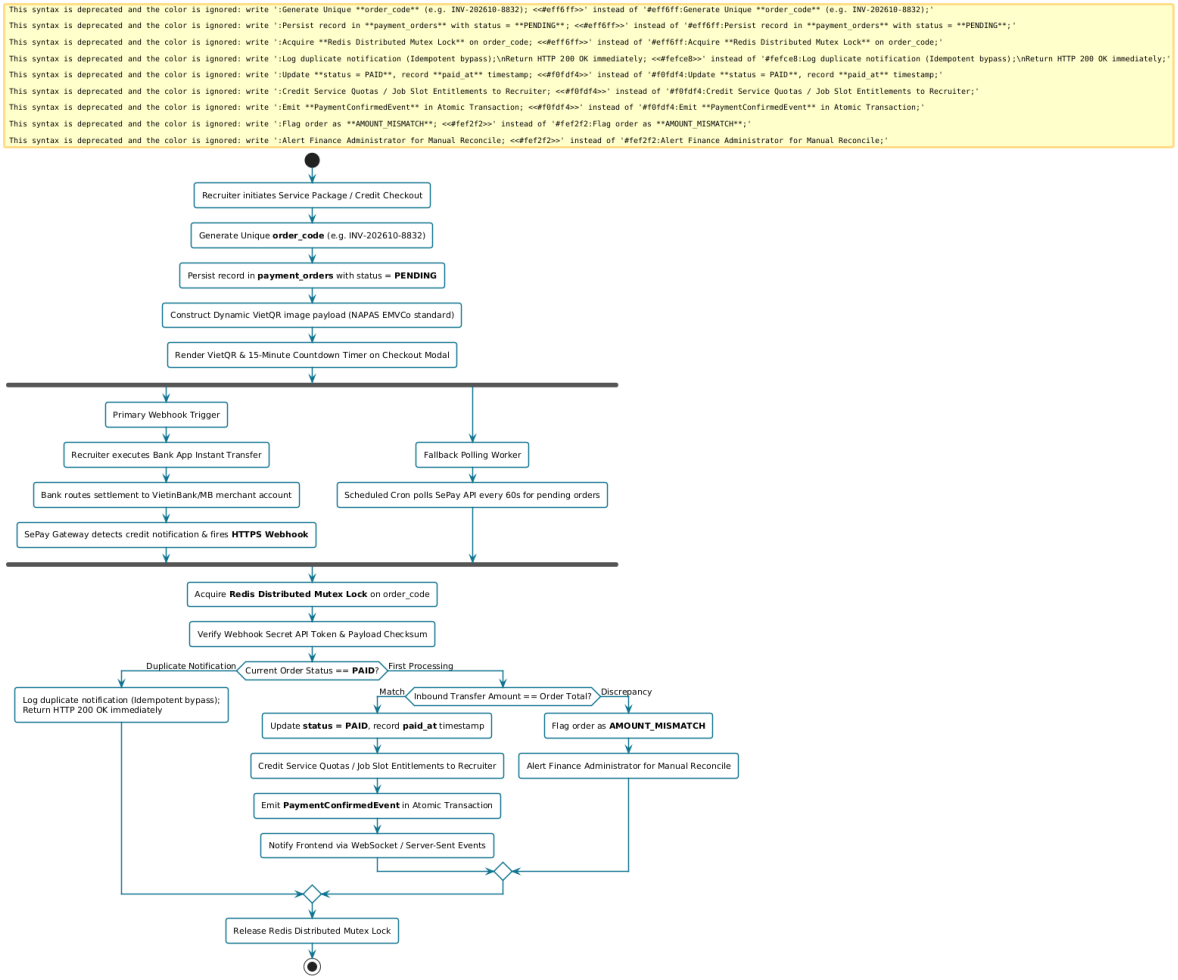


Figure 8.3. Idempotent VietQR Payment Settlement & Dual-Track Reconciliation Flowchart

Acknowledgments and Declaration

Project Documentation for InterVue

Completed on November 11, 2026

The documentation for the **InterVue** project was completed on November 11, 2026. Our team would like to express our sincere gratitude to our supervisor, **Mr. HOÀNG ĐỨC QUANG**, for his continuous guidance, valuable feedback, and encouragement throughout the development of this project.

We are also thankful to our classmates for their constructive input, and to **FPT Aptech** for providing us with the opportunity to apply the knowledge and skills gained during our studies in a practical and meaningful context.

This document is the result of the collective efforts of our team. We confirm that the content presented is original and accurate, with no unauthorized copying or plagiarism. External materials were used for reference only, and any content drafted with the support of digital tools, including artificial intelligence, has been reviewed and verified by our team. We are committed to upholding academic integrity and take full responsibility for this work.

This documentation reflects the dedication and learning process of our team as students still developing our professional skills. We are deeply grateful to our teacher for his support and guidance during this journey.

This is a complete documentation that reflects the dedication, research, and learning process of our team as students who are still developing our professional skills. We sincerely welcome feedback to further improve this

work, and we are deeply grateful to our teacher for his support and guidance during this journey.